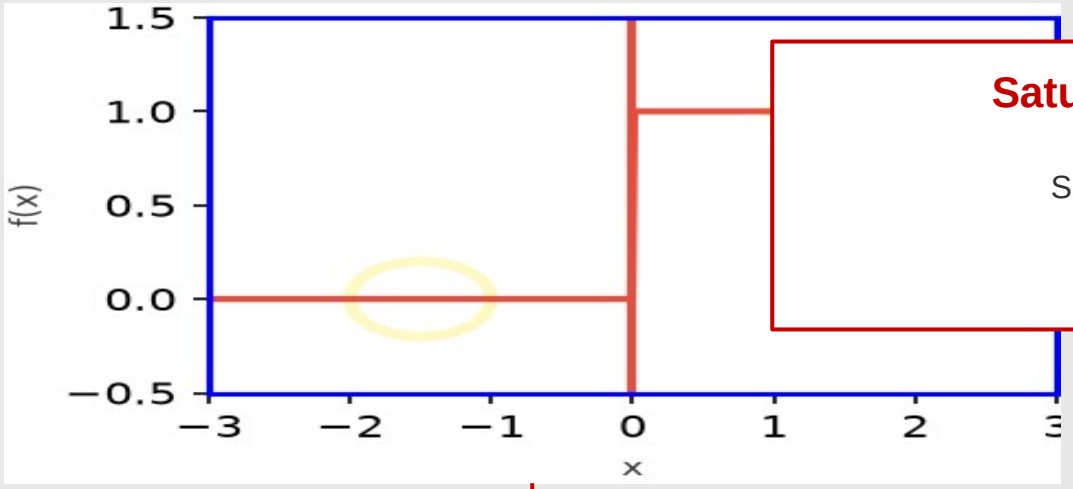
Saturation Problem in Activation Functions

F(x) = Uni-polar Unit Step

F(x) = { 0 for x < 0

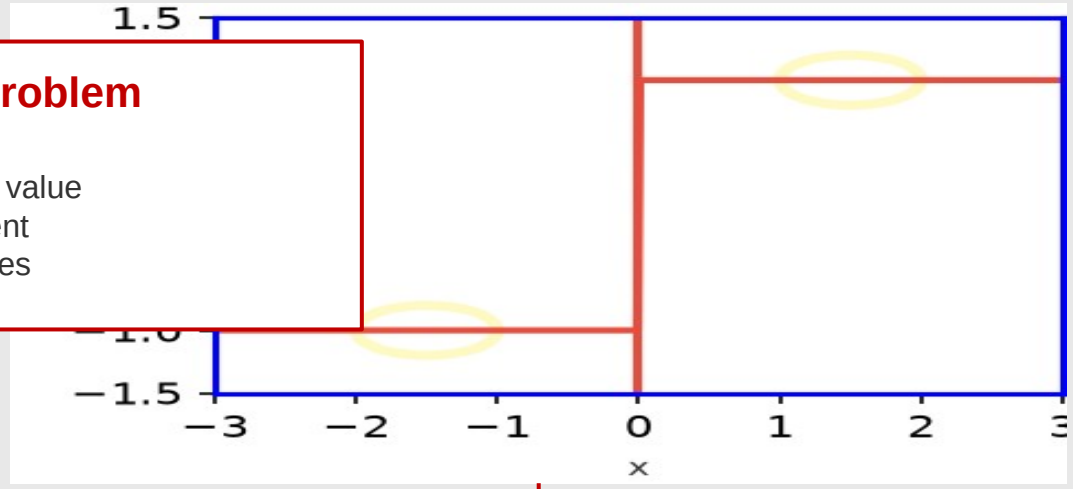
{ 1 for x ≥ 0



F(x) = Bi-polar Unit Step

F(x) = { -1 for x < 0

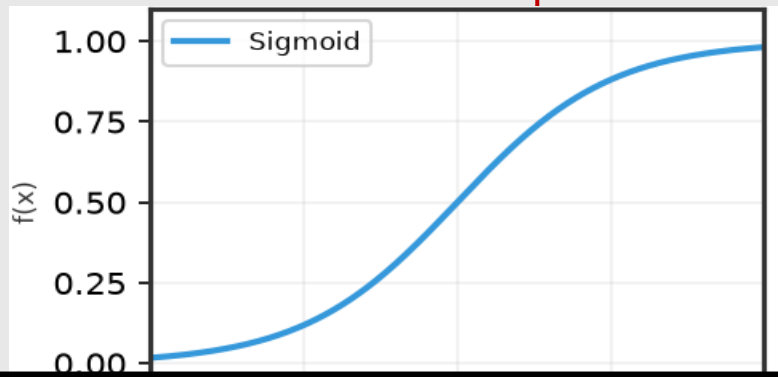
{ 1 for x ≥ 0



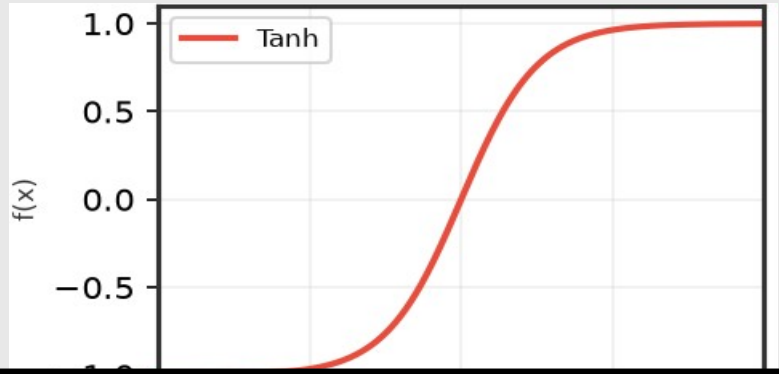
Saturation Problem

Same Output value
for Different
input values

f(x) = Sigmoid(x)



f(x) = tanh(x)



Inverse Tangent Graph

